Writing Assignment #12: Bijective Functions and Permutations

Robert T Kohn

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Question 1

Part B

Claim: $f : (2, \infty) \to (-\infty, -1)$ given by f(x)=$\frac{-x}{x-2}$ is bijective

One to One:
Suppose $\exists a, b \in (2, \infty) \ni f(a) = f(b)$
Then, $\frac{-a}{a-2} = \frac{-b}{b-2}$
Thus, -a(b-2)=-b(a-2)
So, -ab+2a=-ab+2b
Hence, 2a=2b
Then, a=b
Thus, f is injective
Onto:
Suppose $\forall y \in (-\infty, -1)$:
Then, y=$\frac{-x}{x-2}$
Thus, x=$\frac{2y}{y+1}$
So, $\exists x \in (2, \infty) \ni x > 2$
Hence, $\frac{2y}{y+1} > 2$
Then, $2y < 2(y + 1)$
Thus, $0 < 2$
So, $(\forall y \in (-\infty, -1))(\exists x \in (2, \infty)) \ni f(x) = y$
Hence, f is surjective
$\therefore f : (2, \infty) \to (-\infty, -1)$ given by f(x)=$\frac{-x}{x-2}$ is bijective

Question 9

Part B

$u = \{(1, 5), (6, 7), (4, 4), (5, 1), (3, 2), (2, 6), (7, 3)\}$
u=(1 5)(2 6 7 3)

Part F

$v = \{(1,2),(2,5),(3,4),(5,1),(4,6),(6,7),(7,3)\}$

v(u(1))=v(5)=1
v(u(2))=v(6)=7
v(u(3))=v(2)=5
v(u(4))=v(4)=6
v(u(5))=v(1)=2
v(u(6))=v(7)=3
v(u(7))=v(3)=4
v $\circ$u = (2 7 4 6 3 5)